

1 Introduction

We study the interpolated stochastic heat and wave equation (ISHWE) for spatial dimension on $\mathbb{R}^d$,

$$\begin{cases} \left(\partial_t^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2} \right) u(t, x) = I_t^\gamma \left[\sigma(u(t, x)) \dot{W}(t, x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0 & \beta \in (0, 1] \\ u(0, \cdot) = u_0, \quad \partial_t u(0, \cdot) = v_0 & \beta \in (1, 2). \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

2 Preliminaries

For any $t \in [0, T]$ and $x \in \mathbb{R}^d$, we write (assuming $\beta \in (1, 2)$)

$$u(t, x) = J_0(t, x) + I(t, x)$$

where

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + \frac{\partial}{\partial t}[Z(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ I(t, x) &= \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy). \end{aligned}$$

The following is for space-time white noise. For colored noise we need $\rho(\lambda) > 0$ and $2\alpha > d$.

3 Space-time White Noise

In this section we study (1.1) for driven by space-time white noise. To establish the existence and uniqueness of random field solutions in the to (1.1), we will need to assume Dalang's condition:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0, \quad (3.1) \quad \boxed{\text{C:Dalang}}$$

which is equivalent to

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta, \quad (3.2) \quad \boxed{\text{E:DalangEqu}}$$

which is further equivalent to

$$\rho(d) > 0 \quad \text{and} \quad d < 2\alpha, \quad (3.3) \quad \boxed{\text{E:DalangEqu}}$$

where

$$\rho(x) := 2\beta + 2\gamma - 1 - \beta x / \alpha. \quad (3.4) \quad \boxed{\text{D:theta}}$$

3.1 Globally Lipschitz Coefficient

In this section, we assume that the coefficient σ is globally Lipschitz and therefore satisfies the following inequality:

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|,$$

where $c(\sigma) = \sigma(0)$ and $L(\sigma)$ is the Lipschitz constant of σ . We assume that $L(\sigma) > 0$. We now state an existence and uniqueness result proven in [4].

Theorem 3.1. *Under (3.1), the SPDE (1.1) has a unique (in the sense of versions) random field solution $\{u(t, x) : (t, x) \in (0, \infty) \times \mathbb{R}^d\}$ if the initial data are such that*

$$\hat{C}_t := \sup_{t \in [0, T], x \in \mathbb{R}^d} |J_0(t, x)| < \infty. \quad (3.5)$$

Moreover, the following statements are true:

1. $(t, x) \mapsto u(t, x)$ is $L^P(\Omega)$ -continuous for all $p \geq 2$,
2. For all even integers $p \geq 2$, all $t \geq 0$ and $x, y \in \mathbb{R}^d$,

$$\|u(t, x)\|_p^2 \leq 2J_0^2(t, x) + \left[\zeta^2 + 2\hat{C}_t^2 \right] \exp(C\lambda^{2/(1-\sigma)}t). \quad (3.6)$$

1and2normYZ

Lemma 3.2. [4, Theorem 4.1 and Lemma 5.5] Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha} = C_{\#} t^{\rho-1},$$

for all $t > 0$, where $\rho = \rho(d)$ and

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}} Z(t, x) dx = t^{\lceil \beta \rceil - 1} = t.$$

Remark 3.3. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 3.2 gives us that $\int_{\mathbb{R}} G^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (3.7)$$

E:SUPidenti

Proposition 3.4. [11, Proposition 3.2] Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$ and consider $\rho = \rho(d)$ which is defined in (3.4). Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (3.8)$$

E:Nbdd

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (3.9)$$

E:Pnormtbdd

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 3.2 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + \frac{\partial}{\partial t}(Z(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (3.7), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \end{aligned} \quad (3.10) \quad \boxed{\text{E: Jbdd}}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t-s, x-y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t-s, x-y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 3.2 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \int_0^t (t-s)^{\rho-1} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t-s, x-y).$$

Now under Dalang's condition (3.3), the first integral above can be calculated as

$$\int_0^t (t-s)^{\rho-1} ds = \frac{t^\rho}{\rho}.$$

By applying Lemma 3.2, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t-s, x-y) = C_\# \int_0^t ds e^{2as} (t-s)^{\rho-1}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \frac{t^\rho}{\rho} + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{2as} (t-s)^{\rho-1}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_\#}{\rho} t^\rho e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{-2a(t-s)} (t-s)^{\rho-1}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\|I(t, x)\|_p^2 e^{-2at} \leq \frac{8pc^2(\sigma)C_\#}{\rho} t^\rho e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^\infty du e^{-2au} (u)^{\rho-1}$$

$$= \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u) \frac{C_{\#}\Gamma(\rho)}{(2a)^{\rho}}$$

where the last line is true if Dalang's condition [4] holds. If we now apply (3.7) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$N_{a,p}(I) \leq c(\sigma) \sqrt{\frac{8pC_{\#}}{\rho}} \left(\frac{\rho}{2ea}\right)^{\rho/2} + L(\sigma)N_{a,p}(u) \sqrt{\frac{8pC_{\#}\Gamma(\rho)}{(2a)^{\rho}}}. \quad (3.11) \quad \boxed{\text{E:SoleMil3.}}$$

To simplify things, we now let

$$K := \max \left\{ \sqrt{\frac{8C_{\#}}{\rho}} \left(\frac{\rho}{2e}\right)^{\rho/2}, \sqrt{\frac{8C_{\#}\Gamma(\rho)}{2^{\rho}}} \right\}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{\rho/2}} (c(\sigma) + L(\sigma)N_{a,p}(u)). \quad (3.12) \quad \boxed{\text{E:Ibdd}}$$

Now by combining (3.10) and (3.12), we see that

$$N_{a,p}(u) \leq \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_{\infty} & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_{\infty} + \|u_0\|_{\infty} & \beta \in (1, 2). \end{cases} \quad (3.13)$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{\rho/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{\rho/2}} N_{a,p}(u). \quad (3.14) \quad \boxed{\text{E:Nbddgen}}$$

It is important to note that if we now restrict ourselves to the case of $\alpha = \beta = 2$, $d = 1$ and $\gamma = 0$, then (3.11) and (3.14) recover [11, (3.9) and (3.10)] respectively. Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{\rho}}{4L^2(\sigma)K^2} \right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{\rho/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{\rho/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (3.14) implies that

$$N_{a,p}(u) \leq 2 \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_{\infty} & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_{\infty} + \|u_0\|_{\infty} & \beta \in (1, 2) \end{cases} + \frac{c(\sigma)}{L(\sigma)}$$

and this is precisely (3.8). Note that (3.9) follows immediately from (3.8). $\square$

We will need to recall the following result proved in [4] in order to prove Proposition 3.7 below.

P:YInc

Proposition 3.5. [2, Proposition 2.1 and 2.2] Suppose that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, and (3.2) holds. Consider $\rho = \rho(d)$ which is defined above in (3.4). Then $Y(t, x) = Y_{\alpha, \beta, \gamma, d}(t, x)$ satisfies the following:

P:YIncCase1

1. Suppose that $0 < \theta < (\Theta - 1) \wedge 2$ where Θ is given in (3.2), $0 < t, r \leq T$ for some $T > 0$, $\beta \leq 1$ and $\gamma \leq \lceil \beta \rceil - \beta$. Then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^\rho + |x-z|^\theta \right).$$

P:YIncCase2

2. If we only assume that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, $0 < t, r \leq T$ for some $T > 0$ and (3.2) holds then for $0 < \theta < (\Theta - 1) \wedge 2$ there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^q + |x-z|^\theta \right),$$

where

$$0 < q < \rho.$$

Remark 3.6. In Proposition 3.5 above, we may often ignore the dependence of C on $\alpha, \beta, \gamma, \nu, d$ and just write $C(\theta, T)$.

P:NormInc

Proposition 3.7. Suppose that $u_0 = 1$ and that $v_0 = 0$ so that $u(t, x) = 1 + I(t, x)$. Let $a > 0$ and suppose that $0 < \theta < (\Theta - 1) \wedge 2$. Consider $\rho = \rho(d)$. Then for any $x, z \in \mathbb{R}$, $p \geq 2$ and for any $T > 0$ with $t \in [0, T]$,

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)], \quad (3.15)$$

E:NormInc_N

where

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{p}L(\sigma),$$

and

1. $0 < q \leq \rho$ under Case 1 of Proposition 3.5,
2. $0 < q < \rho$ under Case 2 of Proposition 3.5.

Moreover, for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right]$$

we have that

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (3.16)$$

E:NormInc_c

Proof. Note that because of the initial conditions, we see that $\|u(t, x) - u(r, z)\|_p = \|I(t, x) - I(r, z)\|_p$. We now apply Burkholder-Davies-Gundy inequality along with the triangle inequality, as was done in the proof of Proposition 3.4, to see that

$$\|I(t, x) - I(r, z)\|_p^2 \leq 4p \left\| \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \sigma^2(u(s, y)) \right\|_{p/2}$$

$$\leq 4p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \|\sigma^2(u(s, y))\|_{p/2}.$$

Now by applying the Lipchitz condition on σ gives us that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \left(c^2(\sigma) + L^2(\sigma) \|u(s, y)\|_p^2 \right) \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \exp(2as) N_{a,p}^2(u) \right] \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right]. \end{aligned}$$

We now apply Proposition 3.5 to see that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \left[c^2(\sigma) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right], \end{aligned}$$

where $C(\theta, T)$ is as in Proposition 3.5. Now by taking the square root of both sides and applying the identity $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$\|I(t, x) - I(r, z)\|_p \leq 2\sqrt{2C(\theta, T)p} \left[c(\sigma) + L(\sigma) \exp(aT) N_{a,p}(u) \right] \left(|t-r|^q + |x-z|^\theta \right)^{1/2}.$$

This proves (3.15). It is clear that (3.16) follows directly from (3.15) and Proposition 3.4. $\square$

The holder continuity in the next proposition needs to be checked. Also I need to change the $[-R, R]$ to $|x| \leq R$ in the following proof so we can consider higher dimension than $d = 1$.

Check the following Holder continuity with [2].

P:Holder

Proposition 3.8. Suppose that u_0 and v_0 are as on Proposition 3.7 above. Then u has a version, still denoted by u , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$ where $0 < \theta < (\Theta - 1) \wedge 2$ and

1. $0 < q \leq \rho$ under Case 1 of Proposition 3.5,
2. $0 < q < \rho$ under Case 2 of Proposition 3.5.

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

where K is the constant from Proposition 3.4, then we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

Proof. Proposition 3.7 implies that

$$\|u(t, x) - u(r, z)\|_p \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)] \left(|t - r|^q + |x - z|^\theta \right)^{1/2}. \quad (3.17)$$

By Kolmogorov's continuity theorem, there exists a version of u , which we denote by u , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

Next, note that by triangle inequality,

$$\begin{aligned} |u(t, x)|^p &\leq 2^{p-1} (\|u_0\|_\infty^p + |u(t, x) - u_0(x)|^p) \\ &\leq 2^{p-1} \left(\|u_0\|_\infty^p + \sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - u_0(x)|^p) \right) \\ &= 2^{p-1} \left(1 + \sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - 1|^p) \right) \end{aligned}$$

Due to the continuity of u on the compact set $[0, T] \times [-R, R]$, there exists a point $(t_0, x_0) \in [0, T] \times [-R, R]$ such that

$$\sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - 1|^p) = |u(t_0, x_0) - 1|^p.$$

Thus, for any $(t, x) \in [0, T] \times [-R, R]$,

$$|u(t, x)|^p \leq 2^{p-1} (1 + |u(t_0, x_0) - 1|^p)$$

which in return implies that

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \leq 2^{p-1} (1 + |u(t_0, x_0) - 1|^p).$$

Now by taking the expectation of both sides and applying Proposition 3.7 gives us

$$\begin{aligned} \mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] &\leq 2^{p-1} + 2^{p-1} C(p, \theta, T)^p \left((2T)^{q/2} + (2R)^{\theta/2} \right)^p [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)]^p \\ &= 2^{p-1} + C(p, \theta, T, R) [\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} N_{a,p}(u)^p]. \end{aligned}$$

Now if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

then by Proposition 3.4, we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

□

Remark 3.9. The Hölder continuity in time proven above in Proposition 3.8 recovers the result proven by Foondun and Nane in [5, Theorem 1.9] if we restrict ourself to the case where $\beta \in (0, 1)$ and $\gamma = 1 - \beta$.

Nick's Question 3.1
Then

Suppose we consider bounded initial conditions u_0 and v_0 .

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1) \\ [Z(t, \cdot) * v_0](x) + \frac{\partial}{\partial t}[Z(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ &=: \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1) \\ [Z(t, \cdot) * v_0](x) + [Z^*(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases}. \end{aligned}$$

We notice that

$$J_0(t, x) - J_0(s, z) = \begin{cases} [Z(t, \cdot) * u_0](x) - [Z(s, \cdot) * u_0](z) & \beta \in (0, 1) \\ [Z(t, \cdot) * v_0](x) - [Z(s, \cdot) * v_0](z) \\ \quad + [Z^*(t, \cdot) * u_0](z) - [Z^*(s, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases}.$$

Lets just examine the $\beta \in (1, 2)$ case since that case deals with both Z and Z^* . We have that

$$|J_0(t, x) - J_0(s, z)| \leq \int_{\mathbb{R}} |Z(t, x - y) - Z(s, z - y)| |v_0(y)| dy + \int_{\mathbb{R}} |Z^*(t, x - y) - Z^*(s, z - y)| |u_0(y)| dy.$$

For simplicity lets look at the time and space increments separately. For time we have that

$$|J_0(t, x) - J_0(s, x)| \leq \int_{\mathbb{R}} |Z(t, x - y) - Z(s, x - y)| |v_0(y)| dy + \int_{\mathbb{R}} |Z^*(t, x - y) - Z^*(s, x - y)| |u_0(y)| dy.$$

Foondun and Nane [5, Proof of Theorem 1.13] stop right and claim that the above integrals are smooth functions in t . Then it seems that they conclude that they are η -Hölder continuous in t with $\eta = \frac{(1-\beta\gamma/\alpha)}{2}$ where γ is the Reisz Kernel exponent. In other words they say that

$$|J_0(t, x) - J_0(s, x)| \leq C|t - s|^\eta.$$

Are smooth functions also η -Hölder continuous for any $\eta > 0$? If we can argue as in [5], then we can relax our initial conditions to be bounded measurable functions u_0 and v_0 . This is also an improvement from Sanz-Sole and Millet [11] since they assume the initial conditions are continuous.

Nick's Question 3.2 Continue of Question 3.1

Suppose that $\beta \in (0, 1]$.

Time Increment: Suppose $0 \leq s < t$. Then,

$$\begin{aligned} |J(t, x) - J(s, x)| &\leq \|u_0\|_\infty \int_{\mathbb{R}^d} |Z(t, x - y) - Z(s, x - y)| dy \\ &= \|u_0\|_\infty |t - s| \int_{\mathbb{R}^d} \left| \frac{\partial}{\partial t} Z(\xi(s, t), x - y) \right| dy \end{aligned} \quad (3.18)$$

E:DZnorm

where $s \leq \xi(s, t) \leq t$. Now we will calculate $\frac{\partial}{\partial t} Z(\xi(s, t), x - y)$.

$$\begin{aligned} \frac{\partial}{\partial t} Z(\xi(s, t), x) &= \frac{\partial}{\partial t} \left(\pi^{-d/2} t^{[\beta]-1} |x|^{-d} H_{2,3}^{2,1} \left(\frac{|x|^\alpha}{2^{\alpha-1} \nu t^\beta} \middle| \begin{matrix} (1, 1) & ([\beta], \beta) \\ (d/2, \alpha/2) & (1, 1) & (1, \alpha/2) \end{matrix} \right) \right) \\ &= \frac{\partial}{\partial t} \left(\pi^{-d/2} |x|^{-d} H_{2,3}^{2,1} \left(\frac{|x|^\alpha}{2^{\alpha-1} \nu t^\beta} \middle| \begin{matrix} (1, 1) & (1, \beta) \\ (d/2, \alpha/2) & (1, 1) & (1, \alpha/2) \end{matrix} \right) \right) \end{aligned}$$

$$= \pi^{-d/2} |x|^{-d} \frac{\partial}{\partial t} \left(H_{2,3}^{2,1} \left(\frac{1}{(ct)^\beta} \begin{vmatrix} (1,1) & (1,\beta) \\ (d/2, \alpha/2) & (1,1) & (1, \alpha/2) \end{vmatrix} \right) \right)$$

where $c = (2^{\alpha-1} \nu / |x|^{-\alpha})^{1/\beta}$. We now apply [8, (2.2.6)] to see that,

$$\begin{aligned} \pi^{-d/2} |x|^{-d} \frac{\partial}{\partial t} H_{2,3}^{2,1} \left(\frac{1}{(ct)^\beta} \begin{vmatrix} (1,1) & (1,\beta) \\ (d/2, \alpha/2) & (1,1) & (1, \alpha/2) \end{vmatrix} \right) \\ = \frac{\pi^{-d/2} |x|^{-d}}{t} H_{3,4}^{3,1} \left(\frac{1}{(ct)^\beta} \begin{vmatrix} (1,1) & (1,\beta) & (0,\beta) \\ (1,\beta) & (d/2, \alpha/2) & (1,1) & (1, \alpha/2) \end{vmatrix} \right) \\ = \frac{\pi^{-d/2} |x|^{-d}}{t} H_{3,4}^{3,1} \left(\frac{1}{(ct)^\beta} \begin{vmatrix} (1,1) & (0,\beta) & (1,\beta) \\ (1,\beta) & (d/2, \alpha/2) & (1,1) & (1, \alpha/2) \end{vmatrix} \right) \\ = \frac{\pi^{-d/2} |x|^{-d}}{t} H_{2,3}^{2,1} \left(\frac{1}{(ct)^\beta} \begin{vmatrix} (1,1) & (0,\beta) \\ (d/2, \alpha/2) & (1,1) & (1, \alpha/2) \end{vmatrix} \right), \end{aligned}$$

where in the last line we used [8, (2.1.2)]. Replacing c above gives us that

$$\begin{aligned} \frac{\partial}{\partial t} Z(\xi(s, t), x) &= \frac{\pi^{-d/2} |x|^{-d}}{t} H_{2,3}^{2,1} \left(\frac{1}{(ct)^\beta} \begin{vmatrix} (1,1) & (0,\beta) \\ (d/2, \alpha/2) & (1,1) & (1, \alpha/2) \end{vmatrix} \right) \\ &= \frac{\pi^{-d/2} |x|^{-d}}{t} H_{2,3}^{2,1} \left(\frac{|x|^\alpha}{2^{\alpha-1} \nu t^\beta} \begin{vmatrix} (1,1) & (0,\beta) \\ (d/2, \alpha/2) & (1,1) & (1, \alpha/2) \end{vmatrix} \right). \end{aligned}$$

We need to calculate (3.18). In order to do this, we can calculate $\mathcal{F} \left(\frac{\partial}{\partial t} Z(\xi(s, t), \cdot) \right) (0)$. However if we examine the proof of [3, Lemma 4.2], then it appears that

$$\mathcal{F} \left(\frac{\partial}{\partial t} Z(\xi(s, t), \cdot) \right) (\xi) = \frac{1}{t} H_{1,2}^{1,1} \left(2^{-1} \nu t^\beta |\xi|^\alpha \begin{vmatrix} (0,1) \\ (0,1) & (1,\beta) \end{vmatrix} \right).$$

This can no longer be written as a Mittag-Leffler function since by [3, (A.19)] we would have

$$\frac{1}{t} H_{1,2}^{1,1} \left(2^{-1} \nu t^\beta |\xi|^\alpha \begin{vmatrix} (0,1) \\ (0,1) & (1,\beta) \end{vmatrix} \right) = \frac{1}{t} E_{\beta,0} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right),$$

and $E_{\alpha,\beta}$ is not defined for $\beta = 0$. So I am unsure of how to calculate (3.18) in this case. I need to show that

$$\mathcal{F} \left(\frac{\partial}{\partial t} Z(\xi(s, t), \cdot) \right) (0) = \frac{1}{t} H_{1,2}^{1,1} \left(0 \begin{vmatrix} (0,1) \\ (0,1) & (1,\beta) \end{vmatrix} \right) < \infty.$$

3.2 Existence of a Global Solution with Superlinear Diffusion Coefficient

We now impose the following condition on σ : As $|x|, |z| \rightarrow \infty$,

$$|\sigma(x) - \sigma(z)| \leq \sigma_2 |x - z| [\ln_+(|x - z|)]^\delta, \quad (3.19)$$

where $\sigma_2, \delta > 0$ and $\ln_+(z) = \ln(z \vee e)$ for $z > 0$. Further, this implies that as $|x| \rightarrow \infty$,

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta, \quad (3.20)$$

where $\sigma_1 = c(\sigma) = \sigma(0)$.

Remark 3.10. Because of the absence of the drift coefficient in (1.1), we may drop [11, Assumption C1].

Theorem 3.11. *Consider the initial conditions $u_0 = 1$ and $v_0 = 0$ and suppose that σ satisfies (3.19). Then for any $M > 0$, there exists a random field solution to (1.1) in $[-M, M]$ denoted as $(u(t, x) : (t, x) \in [0, T] \times [-M, M])$. This solution is unique and satisfies*

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)| < \infty, a.s. \quad (3.21)$$

Proof. We first solve a *truncated* version of (1.1). By this, we mean (1.1) with σ replaced by σ_N where

$$\sigma_N(x) = \sigma(x) \mathbf{1}_{\{|x| \leq N\}} + \sigma(N) \mathbf{1}_{\{x > N\}} + \sigma(-N) \mathbf{1}_{\{x \leq -N\}}.$$

Note that since σ satisfies (3.19), then σ_N is Lipschitz continuous with Lipschitz constant $L(\sigma_N) := \sigma_2 \ln(2N)^\delta$. In other words,

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|.$$

Hence we can apply Theorem 3.1 and see that there exists a unique solution of the truncated version of (1.1), which we denote by $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$. In addition, Proposition 3.8 implies that the solution has a version, still denoted by u_N , which is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

We will now apply Proposition 3.8 to find an upper bound on the p -norm of u_N . For this, consider

$$a > (8L^2(\sigma_N)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma_N)K^2}\right].$$

Then Proposition 3.8 implies that

$$\mathbb{E} \left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p \right) \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p(N) \exp(apT) \left(2 + \frac{\sigma_1}{L(\sigma_N)} \right) \right] \quad (3.22)$$

where we recall that

$$\mathcal{M}_1 = \sqrt{p}\sigma_1 \quad \text{and} \quad \mathcal{M}_2(N) = \sqrt{pL(\sigma_N)} = \sqrt{p\sigma_2 \ln(2N)^\delta}. \quad (3.23)$$

We now use the above to prove the existence and uniqueness of (1.1) with superlinear σ satisfying (3.19). For any $T > 0$ and $N \geq 2$ with $N \in \mathbb{N}$, we define the following stopping time:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T. \quad (3.24)$$

The uniqueness of the solution and the [local property of the stochastic integral](#) imply that on $\{t < \tau_N\}$, $u_N(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. Hence, $(\tau_N)_{N \geq 2}$ is increasing and bounded above by T .

We now momentarily assume that $\sup_N \tau_N = T$ which would imply that $\{t < \tau_N\} \uparrow \Omega$. On each $\{t < \tau_N\}$, define $(u(t, x) : (t, x) \in [0, T] \times \mathbb{R})$ by $u(t, x) = u_N(t, x)$ and hence $u(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. This implies that on $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on $\{t < \tau_N\}$, we have that $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$ and so u satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since $\{t < \tau_N\} \uparrow \Omega$, we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T] \times \mathbb{R}. \quad (3.25)$$

E:SupLinMil

It is mentioned that the stochastic integrals are no longer defined in $L^2(\Omega)$. The extension of the integral is found in Dalang and Sanz-Sole's textbook that is not yet published.

We now focus on proving that $\sup_N \tau_N = T$, or equivalently that $\mathbb{P}(\tau_N < T) \rightarrow 0$ as $N \rightarrow \infty$. To do this, note that by Chebychev's inequality

$$\begin{aligned} \mathbb{P}(\tau_N < T) &\leq \mathbb{P}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)| \geq N\right) \\ &\leq N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p\right). \end{aligned}$$

An application of (3.22) gives

$$\begin{aligned} N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p\right) &\leq N^{-p} \left(2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p\right]\right) \\ &= \frac{2^{p-1} + C(p, \theta, T, R) \mathcal{M}_1^p}{N^p} + \frac{C(p, T, R) \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p}. \end{aligned}$$

The first term clearly converges to 0 as $N \rightarrow \infty$. As for the second term, we expand using (3.23) to see that

$$\begin{aligned} \frac{\mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p} &= \frac{\left(\sqrt{p\sigma_2 \ln(2N)^\delta}\right)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{\sqrt{p\sigma_2 \ln(2N)^\delta}}\right)^p}{N^p} \\ &\sim \frac{\ln(2N)^{p\delta/2}}{N^p} \rightarrow \infty, \end{aligned}$$

where above we use the symbol $\sim$ to mean asymptotically equivalent as $N \rightarrow \infty$. This completes the proof of the Theorem. $\square$

Remark 3.12. A potential project would be to show that the solution to (1.1) with superlinear coefficient is Holder continuous. This requires to one to extend the Kolmogorov theorem to the type of stochastic integral used to define the mild solution in (3.25). However these stochastic integrals are no longer defined in $L^2(\Omega)$. Maybe a completely new theory needs to be developed to show continuity. [9, Theorem 5.5.5] may be helpful?

4 Spatially Colored Noise Driven SPDE

In this section we will study (1.1) driven by a Gaussian noise that is white in time and colored in space. Informally this means that

$$\mathbb{E}\left[\dot{F}(t, x) \dot{F}(s, y)\right] = \delta_0(t-s) f(x-y)$$

where f is the Riesz kernel with parameter $\lambda \in (0, d)$:

$$f(x) := \frac{1}{|x|^\lambda} \quad \text{and} \quad \mathcal{F}f(\xi) = \frac{K_f}{|\xi|^{d-\lambda}},$$

where

$$K_f = \pi^{-d/2} 2^{-\lambda} \frac{\Gamma((d-\lambda)/2)}{\Gamma(\lambda/2)}.$$

4.1 Lipschitz Diffusion Coefficient

We first examine (1.1) with Lipschitz diffusion coefficient . As we will see below in Theorem 4.2, to establish the existence and uniqueness of a random field solution to (1.1), we will need to assume Dalang's condition:

$$\int_0^t ds \int_{\mathbb{R}^d} dy_1 dy_2 Y(s, y_1) f(y_1 - y_2) Y(s, y_2) < \infty, \quad \text{for all } t > 0, \quad (4.1) \quad \text{C: DalangCl}$$

which is equivalent to

$$\rho(\lambda) > 0 \quad \text{and} \quad \lambda < 2\alpha, \quad (4.2) \quad \text{E: DalangEqu}$$

where we recall from (3.4) that

$$\rho(x) := 2\beta + 2\gamma - 1 - \beta x/\alpha. \quad (4.3)$$

We assume that there exists some $L_\sigma > 0$ such that

$$|\sigma(x) - \sigma(y)| \leq L_\sigma |x - y|.$$

To prove existence and uniqueness of the solution, we will need the following lemma. The following lemma is proven by Foondun and Nane [5] in the case that

$$\beta \in (0, 1) \quad \text{and} \quad \gamma = 1 - \beta \quad (4.4) \quad \text{A: Y=GAssump}$$

Their proof relies on a probabilistic representation of the Green's function for the solution of (1.1). Our proof takes a different approach and relies on the asymptotic behavior of the Euler-Mittag function given in [10, Theorem 1.6]. One should note that under the assumptions (4.4) that $Y(t, x) = Z(t, x)$. However, in general, this is not the case.

Lemma 4.1. *Suppose that $\lambda < 2\alpha$, then there exists a constant $C(\beta, \gamma) =: C > 0$ such that for any $x, y \in \mathbb{R}^d$ we have that*

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) dz dw \leq C K_f \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha}. \quad (4.5) \quad \text{E: YYfBdd}$$

Proof. Through a change of variables, we may write the left side of (4.5) as the following:

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) dz dw = \int_{\mathbb{R}^d} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) du,$$

where $*$ denotes the spatial convolution. Now an application of Parseval's identity gives us that

$$\int_{\mathbb{R}^d} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) du = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x - y))(\xi) \mathcal{F}(Y(t, \cdot) * f)(\xi) d\xi.$$

Applying the translation property of the Fourier transform and [4, Theorem 4.1] gives us that

$$\begin{aligned} & \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x - y))(\xi) \mathcal{F}(Y(t, \cdot) * f)(\xi) d\xi \\ &= \frac{K_f t^{2(\beta\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \exp(-i(x - y)\xi) \frac{E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2}{|\xi|^{d-\lambda}} d\xi \\ &\leq \frac{K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2}{|\xi|^{d-\lambda}} d\xi \end{aligned}$$

where the last inequality follows from the non-negativity of the Euler-Mittag function. Now, by applying [10, Theorem 1.6], we see that there exists a $C > 0$ such that

$$\frac{K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{E_{\beta,\beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2}{|\xi|^{d-\lambda}} d\xi \leq \frac{C K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{1}{\left(1 + \frac{\nu}{2} t^\beta |\xi|^\alpha\right)^2 |\xi|^{d-\lambda}} d\xi.$$

Another change of variables implies that

$$\frac{C K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{1}{\left(1 + \frac{\nu}{2} t^\beta |\xi|^\alpha\right)^2 |\xi|^{d-\lambda}} d\xi \leq \frac{C K_f}{(2\pi)^d} \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{1}{(1 + |\xi|^\alpha)^2 |\xi|^{d-\lambda}} d\xi.$$

We see that the above integral is finite if $\lambda < 2\alpha$ and when this is the case, we have (4.5). $\square$

Nick's Question 4.1 Note that we have $\lambda < 2\alpha$ while Foondun and Nane [5] have that $\lambda < \alpha$. This may be something to look at.

We can now prove the existence and uniqueness of the solution to (1.1) driven by the noise $\dot{F}$.

ExisAndUniq

Theorem 4.2. Suppose that $\lambda\beta/\alpha - 2(\beta + \gamma - 1) < 1$ and in addition, suppose that $\lambda < 2\alpha$. Then there exists a unique $L^2(\Omega)$ -solution to (1.1) driven by $\dot{F}$ such that

$$u(t, x) = J_0(t, x) + I(t, x), \quad a.s.$$

Proof. The proof follows the strategy laid out in Walsh [12]. We start by defining

$$u_0(t, x) := J_0(t, x) \quad \text{and} \quad u_n(t, x) := J_0(t, x) + \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u_n(s, y)) F(ds, dy).$$

In addition, we define

$$D_n(t, x) := \mathbb{E} |u_{n+1}(t, x) - u_n(t, x)|^2 \quad \text{and} \quad H_n(t) := \sup_{x \in \mathbb{R}^d} D_n(t, x).$$

Lastly, we define

$$\Sigma(t, x, n) := |\sigma(u_n(t, x)) - \sigma(u_{n-1}(t, x))|.$$

With this notation we see that

$$D_n(t, x) = \int_0^t ds \int_{\mathbb{R}^d} dz dy Y(t-s, x-y) Y(t-s, x-z) f(y-z) \mathbb{E} (\Sigma(s, y, n) \Sigma(s, z, n)).$$

Now note that by the Lipschitz property of σ , we have that

$$\begin{aligned} \mathbb{E} (\Sigma(s, y, n) \Sigma(s, z, n)) &\leq L_\sigma^2 \mathbb{E} (|u_n(s, y) - u_{n-1}(s, y)| |u_n(s, z) - u_{n-1}(s, z)|) \\ &\leq L_\sigma^2 \|u_n(s, y) - u_{n-1}(s, y)\| \|u_n(s, z) - u_{n-1}(s, z)\| \\ &\leq L_\sigma^2 \sqrt{D_{n-1}(s, y) D_{n-1}(s, z)} \\ &\leq L_\sigma^2 H_{n-1}(s). \end{aligned}$$

Thus,

$$\begin{aligned} D_n(t, x) &\leq L_\sigma^2 \int_0^t H_{n-1}(s) \int_{\mathbb{R}^d} dz dy Y(t-s, x-y) Y(t-s, x-z) f(y-z) \\ &\leq C L_\sigma^2 \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds \frac{H_{n-1}(s)}{(t-s)^{\lambda\beta/\alpha-2(\beta+\gamma-1)}}, \end{aligned}$$

where the last inequality follows from (4.5). In return, this implies that

$$H_n(t) \leq CL_\sigma^2 \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds \frac{H_{n-1}(s)}{(t-s)^{\lambda\beta/\alpha-2(\beta+\gamma-1)}}.$$

Since we assume that $\lambda\beta/\alpha - 2(\beta + \gamma - 1) < 1$, then the result follows from the discussion before the proof of [12, Lemma 3.3]. $\square$

Remark 4.3. One should note that when considering the SHE ($\beta = 1$, $\alpha = 2$ and $\gamma = 0$), the assumptions $\lambda\beta/\alpha - 2(\beta + \gamma - 1) < 1$ and $\lambda < 2\alpha$ reduce down to $\lambda < 2$ and $\lambda < 3$ which recovers the well-known condition that one must have $\lambda < \min\{2, d\}$.

Nick's Question 4.2 In the existence and uniqueness proof in [5, Theorem 1.10], the following is shown

$$H_n(t) \leq \text{const.} \int_0^t \frac{H_{n-1}(s)}{(t-s)^{\gamma\beta/\alpha}}.$$

Note that in this formula, γ is denoting the Riesz kernel parameter. I believe that we need to assume that

$$\gamma\beta/\alpha < 1$$

however in this proof they say we must assume that $d\beta/\alpha < 1$. I think this is a typo. However, this assumption still is not stated in the statement of the theorem. Another issue is that they need to assume that

$$\gamma < \alpha$$

in order to apply [5, Lemma 2.7] and this assumption is not stated in the statement of the theorem. This assumption however would imply that $\gamma\beta/\alpha < 1$ since in [5] $\beta \in (0, 1)$.

In addition, I believe that there is a typo in the statement of [12, Lemma 3.3]. I believe that in Lemma 3.3 *ibid.* we should assume that $a > -1$ and not $a > 1$. This would be consistent with [12, (3.14)].

Nick's Question 4.3 The following tries to give an upper bound to $\sup_{x \in \mathbb{R}^d} \mathbb{E}|u(t, x)|^2$ by using techniques from [5]. There is an issue with the upper bound of $J_0(t, x)$ depending on t when the initial condition $v_0 \neq 0$.

Theorem 4.4 ([5, Theorem 1.10]). *Under the assumptions of Theorem 4.2 we have that*

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}|u(t, x)|^2 \leq \dots \quad (4.6)$$

Proof. The proof follows the strategy from the proof of [5, Theorem 1.10]. We start by defining $A(t) = \sup_{x \in \mathbb{R}^d} \mathbb{E}|u(t, x)|^2$. We will prove the following inequality:

$$A(t) \leq C_1 + C_2 t^{2(\lceil \beta \rceil - 1)} + C_3 \int_0^t \frac{A(s)}{(t-s)^{\beta\lambda/\alpha-2(\beta+\gamma-1)}} ds, \quad (4.7) \quad \boxed{\text{E:Renewal}}$$

To do this, we note that

$$\begin{aligned} \mathbb{E}|u(t, x)|^2 &\leq J_0^2(t, x) + \int_0^t ds \int_{\mathbb{R}^{2d}} dy dz Y(t-s, x-y) Y(t-s, x-z) f(y-z) \mathbb{E}(|\sigma(u(s, y))| |\sigma(u(s, z))|) \\ &:= J_0^2(t, x) + I_2(t, x) \end{aligned} \quad (4.8) \quad \boxed{\text{E:u2normbdd}}$$

Note that for bounded u_0 and v_0 we have that

$$\begin{aligned}
|J_0(t, x)| &\leq \begin{cases} |(Z(t, \cdot) * u_0)(x) & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + |(Z^*(t, \cdot) * u_0)(x) & \beta \in (1, 2) \end{cases} \\
&\leq \begin{cases} \|u_0\|_\infty |(Z(t, \cdot) * 1)(x) & \beta \in (0, 1] \\ \|v_0\|_\infty |(Z(t, \cdot) * 1)(x) + \|u_0\|_\infty |(Z^*(t, \cdot) * 1)(x) & \beta \in (1, 2) \end{cases} \\
&= \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ \|v_0\|_\infty t + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}
\end{aligned}$$

Next we note that

$$\begin{aligned}
\mathbb{E}(|\sigma(u(s, y))||\sigma(u(s, z))|) &\leq L_\sigma^2 \mathbb{E}(|u(s, y)||u(s, z)|) \\
&\leq L_\sigma^2 \mathbb{E}(|u(s, z)|)^{1/2} \mathbb{E}(|u(s, y)|)^{1/2} \\
&\leq A(s).
\end{aligned}$$

Therefore, we can bound I_2 in (4.8) by the following:

$$I_2(t, x) \leq L_\sigma^2 \int_0^t ds A(s) \int_{\mathbb{R}^{2d}} dy dz Y(t-s, x-y) Y(t-s, x-z) f(y-z).$$

Now applying Lemma 4.1 we see that

$$I_2(t, x) \leq C_2 \int_0^t \frac{A(s)}{t^{\beta\lambda/\alpha-2(\beta+\gamma-1)}}.$$

Combining the above gives us (4.7). $\square$

Remark 4.5. The bound in (4.7) is different from [5, Proposition 2.12] for the case of $\beta \in (1, 2)$. The $C_2 t$ term changes things. A similar exponential bound may be found in this case by starting from (4.8) and then using the Piccard iteration techniques from [1, Lemma 2.2].

Proposition 4.6 ([5, Proposition 2.12]). *Let $k > 0$ and suppose that f is locally integrable. Suppose there exists a $K > 0$ satisfies*

$$f(t) \leq K(1+t^2) + K \int_0^t (t-s)^{k-1} f(s) ds. \quad (4.9)$$

E:fRenew

Then

$$f(t) \leq K_1 \exp(K_2 \cdots t). \quad (4.10)$$

Proof. We follow the proof from [6, Proposition 2.5]. For notation, let $g(t) = K(1+t^2)$ and define

$$(\mathcal{A}\theta)(t) := K \int_0^t (t-s)^{k-1} \theta(s) ds$$

for any locally integrable function θ . Then (4.9) can be written as

$$f(t) \leq g(t) + \mathcal{A}\theta(t).$$

By iterating this inequality, we see that

$$f(t) \leq \sum_{i=1}^{n-1} (\mathcal{A}^i g)(t) + (\mathcal{A}^n f)(t).$$

It is stated in the proof of [6, Proposition 2.5] that

$$(\mathcal{A}^n f)(t) = \frac{(K\Gamma(\rho))^2}{\Gamma(n\rho)} \int_0^t (t-s)^{n\rho-1} f(s) ds.$$

Using this, we can calculate $(\mathcal{A}^i g)(t)$ as follows:

$$\begin{aligned} (\mathcal{A}^i g)(t) &= \frac{(K\Gamma(\rho))^i}{\Gamma(i\rho)} \int_0^t (t-s)^{i\rho-1} g(s) ds \\ &= K \frac{(K\Gamma(\rho))^i}{\Gamma(i\rho)} \int_0^t (t-s)^{i\rho-1} (1+s^2) ds \\ &= K \frac{(K\Gamma(\rho))^i}{\Gamma(i\rho)} \cdot \frac{t^{\rho i} (1 + \rho i + t)}{\rho i (1 + \rho i)}. \end{aligned}$$

In addition, it is stated in the proof of [6, Proposition 2.5] that $(\mathcal{A}^n f)(t) \rightarrow 0$ as $n \rightarrow \infty$. Thus, we can say

$$\begin{aligned} f(t) &\leq \sum_{i=0}^{\infty} K \frac{(K\Gamma(\rho))^i}{\Gamma(i\rho)} \cdot \frac{t^{\rho i} (1 + \rho i + t)}{\rho i (1 + \rho i)} \\ &= \sum_{i=0}^{\infty} K \frac{(K t^{\rho} \Gamma(\rho))^i}{\Gamma(i\rho)} \cdot \frac{1 + \rho i + t}{\rho i (1 + \rho i)}. \end{aligned}$$

In the proof of [6, Proposition 2.5], it is stated that

$$\sum_{i=0}^{\infty} \frac{(K t^{\rho} \Gamma(\rho))^i}{\Gamma(i\rho)} = E_{\rho}(K\Gamma(\rho)t^{\rho}),$$

where E_{ρ} is the one-parameter Euler-Mittag function. The $\frac{1+\rho i+t}{\rho i(1+\rho i)}$ term now makes this untrue. $\square$

The following $L^2(\mathbb{R}^d)$ norms may be helpful. I may remove this. Not sure if I actually need it.

2NormZandZ*

Lemma 4.7. Suppose that $d < 2\alpha$ and that $\beta \in (0, 2)$ and that $\gamma \leq 0$. Then,

$$\int_{\mathbb{R}^d} Z^2(t, x) dx = C_Z t^{\lceil \beta \rceil - 1 - d\beta/\alpha} \quad (4.11) \quad \text{E:2NormZ}$$

where $C_Z = (2\pi)^{-d} (\nu/2)^{-d/\alpha} \int_{\mathbb{R}^d} E_{\beta, \lceil \beta \rceil} (-2^{-1} \nu t^{\beta} |\xi|^{\alpha})^2 d\xi$. In addition, we have that

$$\int_{\mathbb{R}^d} Z^{*2}(t, x) dx = C_{Z^*} t^{-d\beta/\alpha} \quad (4.12) \quad \text{E:2NormZ*}$$

where $C_{Z^*} = (2\pi)^{-d} (\nu/2)^{-d/\alpha} \int_{\mathbb{R}^d} E_{\beta, 1} (-2^{-1} \nu t^{\beta} |\xi|^{\alpha})^2 d\xi$.

Proof. We apply Parseval's identity and [4, Theorem 4.1] to see that

$$\int_{\mathbb{R}^d} Z^2(t, x) dx = \frac{t^{\lceil \beta \rceil - 1}}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta, \lceil \beta \rceil} (-2^{-1} \nu t^{\beta} |\xi|^{\alpha})^2 d\xi$$

$$= \frac{t^{\lceil\beta\rceil-1-d\beta/\alpha}}{(2\pi)^d} \left(\frac{\nu}{2}\right)^{-d/\alpha} \int_{\mathbb{R}^d} E_{\beta,\lceil\beta\rceil} (|\xi|^\alpha)^2 d\xi,$$

where in the last line we applied a change of variables. Note that by [10, Theorem 1.6], the integral above on the right hand side is finite when $2\alpha > d$. This proves (4.11). Similarly we have

$$\begin{aligned} \int_{\mathbb{R}^d} Z^{*2}(t, x) dx &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta,1} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right)^2 d\xi \\ &= \frac{t^{-d\beta/\alpha}}{(2\pi)^d} \left(\frac{\nu}{2}\right)^{-d/\alpha} \int_{\mathbb{R}^d} E_{\beta,1} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right)^2 d\xi, \end{aligned}$$

where again the integral above on the right hand side is finite when $d < 2\alpha$. $\square$

4.2 Section 4.2 Millet and Sanz-Solé [11]

Proposition 4.8. [11, Proposition 4.4] *Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$ and for this proof consider $\rho = \rho(\lambda)$ which is defined in (3.4). Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d, \lambda) > 0$ such that for any*

$$a^\rho > (8L^2(\sigma)K^2).$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (4.13) \quad \boxed{\text{E:ColorNbdd}}$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil\beta\rceil - 1)^{\lceil\beta\rceil-1} (ea)^{1-\lceil\beta\rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil\beta\rceil - 1)^{\lceil\beta\rceil-1} (ea)^{1-\lceil\beta\rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (4.14) \quad \boxed{\text{E:ColorPMom}}$$

Proof. We note that $\mathcal{N}_{a,p}(J)$ has the same upper bound as in Proposition 3.4 above. In other words,

$$\mathcal{N}_{a,p}(J) \leq \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

We now work to find an upper bound for $\mathcal{N}_{a,p}(I)$. The Burkholder-Davies-Gundy inequality and Hölder's inequality imply that

$$\|I(t, x)\|_p^2 \leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \|\sigma(u(s, y_1))\|_p \|\sigma(u(s, y_2))\|_p.$$

We now apply the Lipschitz property to see that for $i \in \{1, 2\}$,

$$\|\sigma(u(s, y_i))\|_p \leq c(\sigma) + L(\sigma) \|u(s, y_i)\|_p.$$

Therefore, we may say that

$$\begin{aligned}
\|I(t, x)\|_p^2 &\leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \\
&\quad \times \prod_{i=1}^2 c(\sigma) + L(\sigma) \|u(s, y_i)\|_p \\
&\leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) [c(\sigma) + L(\sigma) \exp(as) N_{a,p}(u)]^2 \\
&\leq 8p c(\sigma)^2 \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \\
&\quad + 8p N_{a,p}(u)^2 L(\sigma)^2 \int_0^t ds \exp(2as) \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2).
\end{aligned}$$

Now by applying Lemma 4.1, we may bound the $\iint_{\mathbb{R}^{2d}} dy_1 dy_2$ integrals by the quantity

$$K_1(t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha},$$

where

$$K_1 := CK_f \left(\frac{\nu}{2}\right)^{-\lambda/\alpha},$$

and C is as in Lemma 4.1. In return, this gives us that

$$\begin{aligned}
\|I(t, x)\|_p^2 &\leq 8p K_1 c(\sigma)^2 \int_0^t (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} ds \\
&\quad + 8p K_1 N_{a,p}(u)^2 L(\sigma)^2 \int_0^t \exp(2as) (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} ds \\
&= 8p K_1 c(\sigma)^2 \frac{t^\rho}{\rho} + 8p K_1 N_{a,p}(u)^2 L(\sigma)^2 \frac{\exp(2at) [\Gamma(\rho) - \Gamma(\rho, 2at)]}{(2a)^\rho} \\
&=: \Delta_1(t) + \Delta_2(t),
\end{aligned}$$

where $\Gamma(\cdot)$ denotes the Gamma function and $\Gamma(\cdot, \cdot)$ is the incomplete-upper Gamma function. Note that

$$\mathcal{N}_{a,p}(I)^2 \leq \sup_{t \geq 0} \exp(-2at) (\Delta_1(t) + \Delta_2(t)).$$

We start with bounding $\sup_{t \geq 0} \exp(-2at) \Delta_1(t)$. Recall the identities in (3.7). With this we see that

$$\sup_{t \geq 0} \exp(-2at) t^\rho = (\rho)^\rho (2ea)^{-\rho}.$$

Hence,

$$\sup_{t \geq 0} \exp(-2at) \Delta_1(t) \leq \frac{8p K_1 c(\sigma)^2}{\rho} \left(\frac{\rho}{2ea}\right)^\rho.$$

Next we bound $\sup_{t \geq 0} \exp(-2at) \Delta_2(t)$. Note that for any $t \geq 0$,

$$\begin{aligned}
\exp(-2at) \Delta_2(t) &= 8p K_1 N_{a,p}(u)^2 L(\sigma)^2 \frac{[\Gamma(\rho) - \Gamma(\rho, 2at)]}{(2a)^\rho} \\
&\leq 8p K_1 N_{a,p}(u)^2 L(\sigma)^2 \frac{\Gamma(\rho)}{(2a)^\rho}.
\end{aligned}$$

Putting this all together gives us that

$$\mathcal{N}_{a,p}(I)^2 \leq \frac{8p K_1 c(\sigma)^2}{\rho} \left(\frac{\rho}{2ea}\right)^\rho + \frac{8p K_1 \Gamma(\rho) L(\sigma)^2}{(2a)^\rho} N_{a,p}(u)^2,$$

and thus,

$$\mathcal{N}_{a,p}(I) \leq 2c(\sigma) \sqrt{\frac{2pK_1}{\rho}} \left(\frac{\rho}{2ea}\right)^{\rho/2} + 2L(\sigma) \frac{\sqrt{2pK_1\Gamma(\rho)}}{(2a)^{\rho/2}} \mathcal{N}_{a,p}(u).$$

Similar to the proof of Proposition 3.4 above, we define

$$K = K(\alpha, \beta, \gamma, d, \lambda) := \max \left\{ 2\sqrt{\frac{2K_1}{\rho}} \left(\frac{\rho}{2e}\right)^{\rho/2}, 2\frac{\sqrt{2K_1\Gamma(\rho)}}{(2)^{\rho/2}} \right\}.$$

We now see that

$$\mathcal{N}_{a,p}(I) \leq Kc(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} + KL(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} \mathcal{N}_{a,p}(u) \quad (4.15)$$

This and the bound for $\mathcal{N}_{a,p}(J)$ found above implies that

$$\begin{aligned} \mathcal{N}_{a,p}(u) &\leq \mathcal{N}_{a,p}(J) + \mathcal{N}_{a,p}(I) \\ &\leq \begin{cases} \|u_0\|_{\infty} & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_{\infty} + \|u_0\|_{\infty} & \beta \in (1, 2) \end{cases} \\ &\quad + Kc(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} + KL(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} \mathcal{N}_{a,p}(u). \end{aligned} \quad (4.16) \quad \boxed{\text{E:N(u)bddCo}}$$

Now consider

$$a^{\rho} > 8K^2L(\sigma)^2.$$

This implies that the interval

$$\left[2, \frac{a^{\rho}}{4K^2L(\sigma)^2} \right]$$

is nonempty. Hence, for any p in this interval we have that

$$\frac{KL(\sigma)\sqrt{p}}{a^{\rho/2}} \leq \frac{1}{2}.$$

This along with (4.16) implies that

$$\begin{aligned} \mathcal{N}_{a,p}(u) &\leq 2 \begin{cases} \|u_0\|_{\infty} & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_{\infty} + \|u_0\|_{\infty} & \beta \in (1, 2) \end{cases} \\ &\quad + \frac{c(\sigma)}{L(\sigma)}. \end{aligned}$$

Its easy to see that (4.14) follows directly from (4.13). □

Proposition 4.9 ([11, Proposition 4.8]). *Statement...*

Proof. By the Burkholder Davies Gundy inequality, we have that

$$\begin{aligned} \|I(t, x) - I(t, z)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 f(y_1 - y_2) \prod_{i=1}^2 (G(t-s, x - y_i) - G(t-s, z - y_i) \sigma(u(s, y_i))) \right\|_{p/2}^2 \\ &=: 4p \|Q\|_{p/2}^2 \end{aligned}$$

We now reorganize the integrand of Q above by following [7, Theorem 3.1]. By expanding the integrand of Q , we get that

$$Q = \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, x - y_1) G(t-s, x - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2))$$

$$\begin{aligned}
& - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, x-y_1) G(t-s, z-y_2) f(y_1-y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\
& - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, z-y_1) G(t-s, x-y_2) f(y_1-y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\
& + \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, z-y_1) G(t-s, z-y_2) f(y_1-y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\
& =: I_1 + I_2 + I_3 + I_4.
\end{aligned}$$

We now introduce the following notation:

$$\begin{aligned}
\Sigma_x(s, y) &= \sigma(u(s, x-y)), \quad \Sigma_{x,z}(s, y) = \sigma(u(s, x-y)) - \sigma(u(s, z-y)), \\
h_1(s, y_1, y_2) &= f(y_1, y_2) \Sigma_{x,z}(s, y_1) \Sigma_{x,z}(s, y_2) \\
h_2(s, y_1, y_2) &= [f(y_1 - y_2 + \xi) - f(y_1 - y_2)] \Sigma_x(s, y_2) \Sigma_{x,z}(s, y_2) \\
h_3(s, y_1, y_2) &= h_2(s, y_2, y_1) \\
h_4(s, y_1, y_2) &= [2f(y_1 - y_2) - f(y_1 - y_2 + \xi) - f(y_1 - y_2 - \xi)] \Sigma_x(s, y) \Sigma_x(s, z).
\end{aligned}$$

With this notation along with a change of variables, we rewrite I_1 through I_4 as the following:

$$\begin{aligned}
I_1 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\
I_2 &= - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2-\xi) \Sigma_x(s, y_1) \Sigma_z(s, y_2) \\
&= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2-\xi) \Sigma_x(s, y_1) \Sigma_{x,z}(s, y_2) \\
&\quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2-\xi) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\
I_3 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2+\xi) \Sigma_x(s, y_2) \Sigma_{x,z}(s, y_1) \\
&\quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2+\xi) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\
I_4 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2) \Sigma_z(s, y_1) \Sigma_z(s, y_2) \\
&= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2) (\Sigma_{x,z}(s, y_1) - \Sigma_x(s, y_1)) (\Sigma_{x,z}(s, y_2) - \Sigma_x(s, y_2)),
\end{aligned}$$

where above I_3 is derived in the same way I_2 was derived. Now if we expand I_4 and regroup all the terms in $I_1 + \dots + I_4$, we get that

$$Q = \sum_{i=1}^4 Q_i \quad (4.17)$$

where

$$Q_i = \int_0^t \iint_{\mathbb{R}^{2d}} dy_1 dy_2 G(t-s, y_1) G(t-s, y_2) f(y_1-y_2) h_i(s, y_1, y_2). \quad (4.18)$$

Thus

$$\|I(t, x) - I(t, z)\|_p^2 \leq 4p \sum_{i=1}^4 \|Q_i\|_{p/2}.$$

We now estimate each $\|Q_i\|_{p/2}$ separately. □

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